Workshop 1

Algebra 2

2025 Semester 1

Warm-up

1. Recall the definitions of:
   - an irreducible element,
   - a prime element,
   - a gcd of two elements a and b.
2. Give an example of an irreducible element in a ring R that is not prime.
3. If d generates the ideal $\langle a, b \rangle$, must d be a gcd of a and b? Is the converse true?
4. What are the prime ideals of $\mathbf{Z}$?
5. Give an example of a UFD that is not a PID.
6. Construct a ring homomorphism $\mathbf{Z}[x] \to \mathbf{Z}/p\mathbf{Z}[y]$ that sends x to y. How many such homomorphisms are there? What do you have to check that the map you defined is a homomorphism? How do you prove that there are no more?

Factorisation in exotic number rings

Let $\omega = e^{2\pi i/3}$ and set $R = \mathbf{Z}[\omega]$. This is the subset of $\mathbf{C}$ consisting of numbers of the form $a + b\omega$, where $a, b \in \mathbf{Z}$. The elements of R are sometimes called *Eisenstein integers*.

1. Check that R is a sub-ring of $\mathbf{C}$.
2. What are the units of $\mathbf{Z}[\omega]$?
3. Factor 3 into irreducibles in $\mathbf{Z}[\omega]$. If you have time, also factor 7. How do you know that your factors are irreducible?
4. Since $\mathbf{Z}[\sqrt{-5}]$ is not a UFD, there must exist an irreducible element that is not prime. Find one.
5. We have seen that $\mathbf{Z}[\sqrt{-5}]$ is not a UFD. This means that it cannot be a PID. Find an ideal that is not principal.

Ideals in polynomial rings

1. Let $\mathbf{Q}[x] \to \mathbf{C}$ be the unique homomorphism that sends x to $\sqrt{2}$. Why is there a unique homomorphism like this? What is its kernel?

2. Let $\mathbf{Q}[x] \to \mathbf{C}$ be the unique homomorphism that sends x to $1/2$. What is its kernel?

3. In both questions above, replace $\mathbf{Q}[x]$ by $\mathbf{Z}[x]$. Must the kernel be principal? Can you find it in these examples?

Factorisation of polynomials

Do this only if you have time. It uses ideas from the readings this week.

1. Find the cubic irreducible polynomials in $\mathbf{F}_2[x]$. Use them to write down cubic irreducible polynomials in $\mathbf{Z}[x]$.

2. Factor $x^4 + 2x^3 - x^2 + 2x - 1$ in $\mathbf{F}_2[x]$ and $\mathbf{F}_3[x]$. What does the factorisation imply about the factorisation in $\mathbf{Z}[x]$.